

Literature Review: Art Gallery Problems for Orthogonal Polygons and Orthogonal Polyhedra

May 22, 2026

Abstract

This review organizes the art-gallery literature most relevant to perimeter-type guard bounds for orthogonal objects. In two dimensions, the perimeter-over-six question is meaningful only after fixing a lattice scale: it is proven for hole-free polyominoes, conditional/open for polyominoes with holes, and open for general integral orthogonal polygons. In three dimensions, the standard term is *orthogonal polyhedron* or *orthogonal polytope*, not “orthonormal.” A direct point-guard analogue of the planar theorem fails: orthogonal polyhedra may require $\Theta(n^{3/2})$ point guards. The natural 3D analogue therefore shifts to edge guards, reflex-edge guards, face guards, or discrete polycube visibility models, where several Urrutia-type conjectures remain open.

1 Reading Guide and Terminology

2D objects. An *orthogonal polygon* is a polygon whose edges are horizontal or vertical. A *polyomino* is a finite edge-connected union of unit axis-parallel squares; its perimeter is naturally the number of unit boundary edges, including hole boundaries when holes are present. An *integral orthogonal polygon* has integer coordinates, so its perimeter is a lattice length. An *ortho-unit polygon* is an orthogonal polygon all of whose edges have length one; in that model, the number of vertices equals the perimeter.

3D objects. The standard 3D term is *orthogonal polyhedron*: a polyhedron whose faces are axis-aligned, equivalently whose face normals are coordinate directions. In higher dimensions, the corresponding term is *orthogonal polytope*. The word *orthonormal* describes vectors or bases, not galleries. Polycubes are the voxel analogue of polyominoes and form a discrete subclass of orthogonal polyhedra.

Guard models. Unless a section says otherwise, a 2D guard is a standard point guard with straight-line visibility inside the polygon. In 3D the model must be stated: the literature distinguishes point guards, vertex guards, edge guards, reflex-edge guards, face guards, open edge guards, and discrete polycube visibility graph models. These models are not interchangeable.

Why perimeter needs a lattice scale. No theorem of the form “at most $P/6$ guards” can hold for arbitrary real-coordinate orthogonal polygons if P is Euclidean perimeter: scaling a polygon down preserves the guard number while making P arbitrarily small. The perimeter-over-six question is therefore a lattice-normalized question: polyomino perimeter, integral perimeter, or a fixed unit-edge model.

2 Flow of the Literature

1. Classical art-gallery theorems start with arbitrary simple polygons: Chvatal proves $\lfloor n/3 \rfloor$ and Fisk gives the standard triangulation proof [7, 11]. Shermer, O’Rourke, and Urrutia provide the main survey/book context [24, 20, 25].
2. Orthogonal polygons admit stronger vertex-count theorems: Kahn–Klawe–Kleitman prove the tight $\lfloor n/4 \rfloor$ bound for simple orthogonal polygons [15].
3. Holes complicate the 2D theory. Vertex-guard conjectures for orthogonal polygons with holes, especially Shermer’s and Hoffmann’s formulations, remain separate from the perimeter-over-six point-guard question [26, 14, 31].
4. Polyominoes introduce area and lattice-perimeter parameters. The tight area theorem allows holes [3], while Maßberg’s perimeter-over-six theorem is proven for hole-free polyominoes and becomes conditional in the presence of holes [17, 18].
5. Recent ortho-unit work proves a strong unit-edge theorem and states the broader integral-perimeter conjecture [8].
6. In 3D, point guards are too weak for a planar-style theorem: orthogonal polyhedra need $\Theta(n^{3/2})$ point guards in the worst case [21, 27]. Edge-guard and reflex-edge-guard conjectures become the main 3D analogue [2, 29, 6].

3 One-Page Status Matrix

Table 1 is the fastest way to read the review. The labels “proven tight,” “conditional,” “conjectural,” and “open” refer to the exact model in the first three columns; changing the guard type, allowing holes, or changing the parameter can change the answer.

Model	Guards	Parameter	Status	Main source
Simple orthogonal polygon	point	vertices n	Proven tight: $\lfloor n/4 \rfloor$	Kahn–Klawe–Kleitman; Gyori [15, 13]
Orthogonal polygon with holes	vertex	vertices n , holes h	Open at the strongest conjectured bounds; partial results for restricted cases	Urrutia; Hoffmann–Kriegel; Zylinski [26, 14, 31]
Hole-free polyomino	point	lattice perimeter ℓ	Proven tight for $\ell \geq 6$: $\lfloor \ell/6 \rfloor$	Maßberg [17]
Polyomino with holes	point	lattice perimeter ℓ	Conditional/open: Maßberg’s extension depends on a rectangle-packing conjecture	Maßberg habilitation [18]
Polyomino with holes	point	cells m	Proven tight: $\lfloor (m+1)/3 \rfloor$; does not imply perimeter-over-six	Biedl et al. [3]
Ortho-unit polygon	point	perimeter $P = n$	Proven tight for $n \geq 12$: $\lfloor (n-4)/8 \rfloor$, stronger than $P/6$ in this model	Diaz-Banez et al. [8]
Integral orthogonal polygon	point	lattice perimeter N	Conjectural: $\lfloor N/6 \rfloor$; no proof found in this search	Diaz-Banez et al. [8]
Orthogonal polyhedron	point	vertices n	No linear planar analogue: worst case is $\Theta(n^{3/2})$	Paterson–Yao; Viglietta [21, 27]
Orthogonal polyhedron	edge or reflex edge	edges m , reflex edges r	Main 3D analogue; Urrutia-type conjectures remain open beyond partial classes	Benbernou et al.; Viglietta; Cano et al. [2, 29, 6]

Table 1: Status map for orthogonal art-gallery problems closest to perimeter-over-six.

4 Proof Techniques and Where They Break

Triangulation and coloring.

Chvatal and Fisk establish the classical template for simple polygons [7, 11]. Kahn–Klawe–Kleitman and Gyori adapt coloring ideas to simple orthogonal polygons [15, 13]. The method is powerful for vertex-count bounds but does not directly produce a lattice-perimeter theorem.

Quadrilateralization and cactus duals.

Orthogonal holes are often attacked by decomposing the domain into rectangles or quadrilaterals and coloring a dual structure. Zylinski’s cactus-dual condition explains why the $\lfloor (n + h)/4 \rfloor$ conjecture is approachable in some cases but not yet resolved in full [31].

Rectangle packing and perfect-graph methods.

Motwani–Raghunathan–Saran, Worman–Keil, and Maßberg connect orthogonal guarding with star polygons, rectangle systems, and perfect graphs [19, 30, 17]. This is the proof technology behind the hole-free polyomino perimeter theorem. Its failure point is explicit: the hole case depends on Maßberg’s maximal-rectangle packing conjecture [18].

Area and grid-cell arguments.

Biedl et al. prove the tight $\lfloor (m + 1)/3 \rfloor$ theorem for polyominoes with holes by using the cell structure rather than perimeter [3]. This is strong evidence that holes are tractable in a discrete model, but it leaves open whether perimeter alone controls the guard number.

Hardness and approximation reductions.

Schuchardt–Hecker, Katz–Roisman, and later work on sliding cameras and transmitters show that optimization versions remain hard even in restricted orthogonal settings [23, 16, 9, 4]. These results do not contradict extremal upper bounds; they warn that finding a minimum guard set is a different problem from proving a universal bound.

3D decomposition and binary-space partitions.

Paterson–Yao’s binary-space-partition theorem, as used in Viglietta’s survey, shows that point guards in orthogonal polyhedra have $\Theta(n^{3/2})$ worst-case behavior [21, 27]. This is the main reason a 3D project should not simply ask for a point-guard analogue of the planar perimeter theorem.

Edge-guard and reflex-edge induction.

Benbernou et al., Viglietta, and Cano–Toth–Urrutia–Viglietta show that the most promising 3D analogue is based on edges, reflex edges, and genus rather than point guards [2, 29, 6]. The remaining gap is structural: the known upper bounds do not meet the Urrutia-type lower-bound constructions.

5 Status by Model

Simple orthogonal polygons, point guards.

Proven tight: every n -vertex simple orthogonal polygon is guardable by $\lfloor n/4 \rfloor$ point guards [15, 13].

Orthogonal polygons with holes, vertex guards.

Several upper bounds and partial sharp results are known. O’Rourke gives the classical $\lfloor (n + 2h)/4 \rfloor$ style bound for h holes [20]. Shermer’s stronger $\lfloor (n + h)/4 \rfloor$ conjecture is open

in general; Zylinski gives a coloring proof of Aggarwal’s theorem for $h \leq 2$ and cactus-dual quadrilateralizations [26, 31].

Hole-free polyominoes, point guards, lattice perimeter ℓ .

Proven tight: for $\ell \geq 6$, Maßberg proves that $\lfloor \ell/6 \rfloor$ point guards suffice [17]. Use $\max\{1, \lfloor \ell/6 \rfloor\}$ if the one-square case is included.

Polyominoes with holes, point guards, lattice perimeter ℓ .

Open/conditional. Maßberg’s text explicitly states that the extension to holes would follow from a maximal-rectangle packing conjecture [18]. No later proof was found in this literature search, current to May 22, 2026.

Polyominoes with holes, point guards, area m .

Proven tight: every m -cell polyomino, holes allowed, is guardable by $\lfloor (m+1)/3 \rfloor$ point guards [3]. This does not settle perimeter-over-six.

Ortho-unit polygons, point guards.

Proven tight: every ortho-unit polygon with $n \geq 12$ vertices is guardable by $\lfloor (n-4)/8 \rfloor$ guards [8]. Since $n = P$ in the ortho-unit model, this is stronger than $P/6$ for that narrow class.

Integral orthogonal polygons, point guards, perimeter N .

Open. Diaz-Banez et al. state the natural conjecture that $\lfloor N/6 \rfloor$ guards suffice [8]. No later proof was found in this literature search, current to May 22, 2026.

Polycubes and polyhypercubes.

Pinciu studies discrete polyform variants, including polycubes and polyhypercubes [22]. These are relevant for 3D lattice analogues, but they use a discrete polyform/visibility-graph framework rather than the continuous point-guard model for orthogonal polyhedra.

Orthogonal polyhedra, point guards.

No planar-style linear theorem exists in terms of vertices. The Paterson–Yao bound gives $\Theta(n^{3/2})$ point guards as sufficient and sometimes necessary for n -vertex orthogonal polyhedra [21, 27].

Orthogonal polyhedra, edge and reflex-edge guards.

The central open direction is an Urrutia-type edge-guard theorem: lower-bound examples require roughly $m/12$ edge guards, while known upper bounds are larger. Benbernou et al. improve the orthogonal upper bound to asymptotic $(11/72)m$ edge guards, or $(7/12)r$ in terms of reflex edges [2]. Viglietta proves optimal-style bounds for 2-reflex orthogonal polyhedra [29]. Cano, Toth, Urrutia, and Viglietta improve the general 3D polyhedron edge-guard bound to $(5/6)m$, while noting a substantial remaining gap [6].

6 Two-Dimensional Orthogonal Polygons

6.1 Classical Background

The original art-gallery theorem is due to Chvatal: $\lfloor n/3 \rfloor$ guards always suffice for a simple n -vertex polygon, and this is tight [7]. Fisk’s short proof via triangulation and three-coloring became the standard proof template [11]. O’Rourke’s book remains the standard comprehensive reference for the classical theory [20]; Shermer and Urrutia survey art-gallery and illumination problems, including variants and open problems [24, 25]. On the algorithmic side, Abrahamsen, Adamaszek,

and Miltzow prove that the general art-gallery problem is $\exists\mathbb{R}$ -complete, underscoring why restricted orthogonal and lattice models are usually treated separately [1].

6.2 Simple Orthogonal Polygons

Kahn, Klawe, and Kleitman prove the tight orthogonal theorem: $\lfloor n/4 \rfloor$ point guards suffice for every simple orthogonal polygon with n vertices [15]. Gyori gives a short proof [13]. This theorem is a vertex-count theorem, not a perimeter theorem. In a lattice setting the vertex count and the perimeter may be very different, which is why the polyomino and integral-perimeter questions are separate.

The proof tradition connects to decomposition, coloring, and perfect-graph methods. Motwani, Raghunathan, and Saran study covering orthogonal polygons with star polygons [19]; Worman and Keil develop decomposition and visibility-graph machinery for orthogonal art-gallery problems [30]. Bjørling-Sachs proves the tight $\lfloor n/4 \rfloor$ edge-guard bound for rectilinear polygons, a useful 2D reference point for later edge-guard models [5].

6.3 Holes

Holes are one of the main fault lines in the literature. For orthogonal polygons with h holes, classical vertex-guard bounds exist, but the strongest natural conjectures are not settled in general. Urrutia’s open-problem list records Shermer’s conjecture that an n -vertex orthogonal polygon with h holes should be guardable by $\lfloor (n + h)/4 \rfloor$ vertex guards, and Hoffmann’s conjecture that $\lfloor 2n/7 \rfloor$ vertex guards should suffice for orthogonal polygons with holes [26]. Zylinski proves the $\lfloor (n + h)/4 \rfloor$ bound under a cactus-dual quadrilateralization condition and recovers Aggarwal’s theorem for $h \leq 2$ [31]. Hoffmann and Kriegel prove graph-coloring consequences for rectilinear polygons with holes, including upper bounds for rectilinear prison-yard variants [14].

These hole results are relevant context, but they do not prove the perimeter-over-six statement. They usually concern vertex guards and vertex counts; the perimeter-over-six problem concerns point guards and lattice perimeter.

7 Polyomino and Perimeter Results

7.1 Hole-Free Polyominoes

Maßberg proves the direct perimeter theorem for hole-free polyominoes [17]. In the notation of that paper, if a hole-free polyomino has perimeter $\ell \geq 6$, then it can be guarded by at most $\lfloor \ell/6 \rfloor$ point guards. The proof uses maximal axis-parallel rectangles in a rectilinear gallery, relates a rectangle packing number to the guard number, and then bounds the packing by the perimeter. Comb-like examples show tightness.

7.2 Polyominoes with Holes

The area theorem does allow holes: Biedl, Irfan, Iwerks, Kim, and Mitchell prove that every m -cell polyomino can be guarded by $\lfloor (m + 1)/3 \rfloor$ point guards, and the bound is tight [3]. This is the strongest clean theorem for arbitrary polyominoes with holes in the standard point-guard model, but it is an area bound rather than a perimeter bound.

For perimeter, the hole case remains conditional. Maßberg observes that the perimeter theorem would extend to holes if a maximal-rectangle packing conjecture were true [18]. The important

point for a new paper is that this is *not* a proof of perimeter-over-six for holes. It identifies the missing structural statement.

7.3 Ortho-Unit and Integral Orthogonal Polygons

Diaz-Banez et al. prove that every ortho-unit polygon with $n \geq 12$ vertices can be guarded by $\lfloor (n-4)/8 \rfloor$ guards, and that the bound is tight [8]. Since every edge has unit length in an ortho-unit polygon, n is the perimeter. This solves a narrow unit-edge class more strongly than $P/6$.

The same paper frames the broader integral-perimeter conjecture: every integral orthogonal polygon of perimeter N should be guardable by $\lfloor N/6 \rfloor$ point guards. This remains the closest published statement of the general perimeter-over-six problem beyond hole-free polyominoes.

8 Three-Dimensional Orthogonal Polyhedra

8.1 Point Guards

The direct 3D point-guard analogue of planar art-gallery theorems fails. In orthogonal polyhedra with n vertices, Paterson and Yao’s binary-space partition result implies a tight $\Theta(n^{3/2})$ point-guard bound [21]. Viglietta’s thesis presents this as Theorem 3.7 and draws the key modeling conclusion: point guards do not yield a linear orthogonal-polyhedron art-gallery theorem [27]. Therefore a 3D “perimeter-over-six” theorem is not obtained by simply replacing polygons with orthogonal polyhedra and keeping point guards.

8.2 Edge and Reflex-Edge Guards

The 3D literature often treats edge guards as the analogue of planar vertex guards. Urrutia conjectured that an orthogonal polyhedron with m edges should be guardable by $m/12 + O(1)$ closed edge guards, matching known lower bound examples up to an additive constant [27]. A related reflex-edge conjecture asks whether roughly $(r-g)/2 + 1$ reflex edge guards suffice, where r is the number of reflex edges and g is the genus.

Benbernou, Demaine, Demaine, Kurdia, O’Rourke, Toussaint, Urrutia, and Viglietta improve the known asymptotic upper bounds for orthogonal polyhedra: $(11/72)m$ edge guards suffice in terms of total edges, and $(7/12)r$ suffice in terms of reflex edges [2]. Viglietta later proves optimal-style bounds for 2-reflex orthogonal polyhedra, namely orthogonal polyhedra whose reflex edges occur in only two directions [29]. These are major partial results, but they do not settle the general 3-reflex orthogonal-polyhedron conjectures.

Cano, Toth, Urrutia, and Viglietta give a broad theorem for arbitrary polyhedra in three-space: every polyhedron with m edges can be guarded by at most $(5/6)m$ edge guards [6]. This improves the general-polyhedron edge-guard bound, but it still leaves a large gap from the conjectured $m/6$ style lower/upper behavior for general polyhedra and the $m/12 + O(1)$ orthogonal-polyhedron conjecture.

8.3 Face Guards and Polycube Variants

Face guards are another 3D model. Viglietta studies face-guarding for several classes of polyhedra, including orthogonal polyhedra, 4-oriented polyhedra, and 2-reflex orthostacks [28]. Face guards are useful for mapping the 3D model space, but they are not a direct continuation of the point-guard perimeter-over-six question.

For lattice 3D objects, Pinciu’s work on polycubes and polyhypercubes gives a discrete polyform line of results [22]. This is the natural place to look if the intended 3D project is voxel/discrete rather than continuous orthogonal-polyhedron visibility.

9 How Open Status Is Classified

This review uses the following labels for open-problem evidence.

Proven tight.

A peer-reviewed theorem gives an upper bound and a matching lower-bound construction in the same model.

Source-stated conjecture.

A paper, thesis, or curated open-problem page explicitly states the question as a conjecture or open problem.

Conditional theorem gap.

A source claims that a desired result follows if another named conjecture or structural lemma is true. The downstream result should be cited as conditional, not as proven.

Derived research direction.

The review identifies a natural next question by comparing known theorems and failures. These are useful project directions, but they should not be cited as named conjectures unless a primary source is later found.

10 Quoted Open-Problem Evidence

The following short quotations are included only to document open status. Each quote is followed by the conclusion that should be drawn from it.

“If this conjecture is true”

— Maßberg, immediately before the claimed hole extension [18]

Status confirmed. Maßberg’s hole extension is conditional on Conjecture 6.10, not proven. The conjecture concerns whether a packing of maximal rectangles bounds the guard number in rectilinear galleries with holes. No later proof was found in this literature search, current to May 22, 2026.

“also true for polyominoes with holes”

— Maßberg, conditional conclusion after Conjecture 6.10 [18]

Status confirmed. This quote must not be used without the preceding condition. The published theorem is hole-free; the hole case is a stated conditional extension.

“guarded with at most $\lfloor N/6 \rfloor$ guards”

— Diaz-Banez et al., integral-perimeter conjecture [8]

Status confirmed. This is a conjecture in the 2025 ortho-unit paper. The paper proves the ortho-unit theorem, not the full integral orthogonal polygon conjecture. No later proof was found in this literature search, current to May 22, 2026.

“Floor $[(n+h)/4]$ vertex guards”

— Urrutia’s page recording Shermer’s orthogonal-holes conjecture [26]

Status confirmed. This is a vertex-guard conjecture for orthogonal polygons with holes. It is relevant to holes, but it is not the same as the point-guard perimeter-over-six question.

“no linear upper bound”

— Viglietta, discussing point guards in orthogonal polyhedra [27]

Status confirmed. This is the main reason 3D point guards should not be treated as a direct analogue of the 2D perimeter theorem.

“Further reducing this gap is left as an open problem.”

— Cano, Toth, Urrutia, and Viglietta, on edge guards in 3-space [6]

Status confirmed. As of the 2022 general edge-guard paper, the gap between lower bounds and upper bounds for edge guarding polyhedra remains an explicit open problem.

“Is there a constant-factor approximation for polyominoes with holes?”

— Filtser et al., for k -hop visibility in polyominoes [10]

Status confirmed. This is a current open problem in a related discrete visibility model, not in standard continuous point-guard visibility.

11 Relevant Open Problems

Problems 1–3 are closest to the perimeter-over-six question. Problems 4–5 are the main 2D orthogonal-with-holes vertex-guard conjectures. Problems 7–10 are the 3D orthogonal-polyhedron directions. Each item records whether its open status is source-stated, conditional, or derived from the literature.

1. Perimeter-over-six for polyominoes with holes.

Kind: conditional theorem gap; priority: high.

Let P be a polyomino with holes and total lattice perimeter ℓ , including all hole boundaries. Is P always guardable by $\max\{1, \lfloor \ell/6 \rfloor\}$ point guards?

Open-status check: confirmed by Maßberg’s conditional wording in the habilitation text; No later proof was found in this literature search, current to May 22, 2026.

2. Maßberg’s maximal-rectangle packing conjecture for holes.

Kind: source-stated conjecture; priority: high.

In a rectilinear gallery with holes, does the maximum packing size of maximal rectangles upper-bound the number of point guards?

Open-status check: source-confirmed as Conjecture 6.10 in Maßberg’s text. A proof would extend the hole-free perimeter theorem to polyominoes with holes.

3. Integral orthogonal polygon perimeter conjecture.

Kind: source-stated conjecture; priority: high.

Is every integral orthogonal polygon of perimeter N guardable by $\lfloor N/6 \rfloor$ point guards?

Open-status check: source-confirmed as the natural conjecture in the 2025 ortho-unit paper; No later proof was found in this literature search, current to May 22, 2026.

4. Shermer’s vertex-guard conjecture for orthogonal polygons with holes.

Kind: source-stated conjecture; priority: medium-high.

Is every n -vertex orthogonal polygon with h holes guardable by $\lfloor (n + h)/4 \rfloor$ vertex guards?

Open-status check: Urrutia’s open-problem list records it as open; Zylinski proves important partial cases, including Aggarwal’s $h \leq 2$ theorem.

5. Hoffmann’s vertex-guard conjecture for orthogonal polygons with holes.

Kind: source-stated conjecture; priority: medium.

Does $\lfloor 2n/7 \rfloor$ always suffice for vertex guarding any orthogonal polygon with holes?

Open-status check: Urrutia’s page records this as a distinct open conjecture. It is vertex-count based and does not imply perimeter-over-six.

6. If perimeter-over-six with holes is false, find the correction term.

Kind: derived research direction; priority: high if counterexamples emerge.

Possible parameters include total perimeter, outer perimeter, number of holes, hole perimeter, reflex vertices, or rectangle-packing obstructions.

Open-status check: this is a derived research direction, not a named source conjecture. It follows from the current gap in the hole case.

7. 3D Urrutia edge-guard conjecture for orthogonal polyhedra.

Kind: source-stated conjecture; priority: high for 3D.

Is every genus-zero orthogonal polyhedron with m edges guardable by $m/12 + O(1)$ edge guards?

Open-status check: recorded in Viglietta’s thesis and subsequent edge-guard literature; partial upper bounds are known, but the general conjecture is not settled.

8. 3D reflex-edge conjecture for orthogonal polyhedra.

Kind: source-stated conjecture with proven restricted cases; priority: high for 3D.

Is every orthogonal polyhedron with $r > 0$ reflex edges and genus g guardable by $\lfloor (r - g)/2 \rfloor + 1$ reflex-edge guards?

Open-status check: Viglietta proves this for 2-reflex orthogonal polyhedra and states the general version as a conjecture.

9. Close the general 3D edge-guard gap.

Kind: source-stated open problem; priority: medium-high.

For arbitrary polyhedra with m edges, improve the current gap between lower-bound examples and the $(5/6)m$ upper bound.

Open-status check: Cano, Toth, Urrutia, and Viglietta explicitly state that reducing this gap remains open.

10. Choose the correct 3D analogue of perimeter.

Kind: derived research direction; priority: high before theorem work.

For orthogonal polyhedra or polycubes, decide whether the right parameter is total edges, reflex edges, surface area, voxel count, genus, or a combination.

Open-status check: not a single named theorem, but forced by the literature: point guards have a superlinear vertex bound, while edge/reflex edge and polycube models have different behavior.

11. Algorithmic complexity and approximation for restricted orthogonal models.

Kind: mixed source-stated and derived direction; priority: medium.

Minimum guarding is NP-hard in many orthogonal settings [23, 16, 3, 9, 4]. Identify exact polynomial subclasses and approximation guarantees for polyominoes with holes, integral orthogonal polygons, polycubes, and orthogonal polyhedra.

Open-status check: several hardness results are known, while recent work on k -hop polyomino visibility still asks for a constant-factor approximation for polyominoes with holes [10].

12 Search Method and Source-Quality Notes

This review was checked on May 22, 2026. The source search prioritized peer-reviewed theorem statements, author-posted PDFs, theses when they contain explicit conjectures or model summaries, and curated open-problem pages by researchers in the area. Searches used combinations of:

- orthogonal art gallery holes;
- polyomino perimeter guards;
- ortho-unit polygons guarded perimeter;
- integral orthogonal polygon perimeter guards;
- orthogonal polyhedra edge guards;
- reflex edge guards 2-reflex orthogonal polyhedra;
- k -hop visibility polyominoes holes.

The review treats sources in the following order of strength:

1. Peer-reviewed theorem sources with exact guard model and parameter.
2. Theses or author PDFs when they state a conjecture, collect a known theorem, or give a precise proof dependency.
3. Curated open-problem pages when they record named conjectures and no later proof was found.
4. Derived questions created by comparing the known theorems and gaps.

The phrase “no later proof was found” is not itself a mathematical proof of openness. It records the outcome of this search and is included to prevent conditional statements from being accidentally cited as established theorems.

13 Journals and Venues to Monitor

Journals

- *Discrete & Computational Geometry*
- *Computational Geometry: Theory and Applications*
- *Journal of Computational Geometry*
- *International Journal of Computational Geometry & Applications*
- *Graphs and Combinatorics*
- *Algorithmica*
- *Journal of the ACM*
- *ACM Transactions on Algorithms*
- *SIAM Journal on Discrete Mathematics*
- *SIAM Journal on Computing*
- *Journal of Computer and System Sciences*
- *Journal of Algorithms* and successor venues
- *The Electronic Journal of Combinatorics*
- *Discrete Applied Mathematics*
- *Information Processing Letters*
- *Journal of Discrete Algorithms*
- *Theoretical Computer Science*
- *Mathematical Logic Quarterly*
- *Electronic Notes in Discrete Mathematics*

Conferences and Workshops

- Symposium on Computational Geometry (SoCG)
- European Workshop on Computational Geometry (EuroCG)
- Canadian Conference on Computational Geometry (CCCG)
- Latin American Symposium on Theoretical Informatics (LATIN)
- International Symposium on Algorithms and Computation (ISAAC)
- European Symposium on Algorithms (ESA)
- ACM-SIAM Symposium on Discrete Algorithms (SODA)
- Workshop on Algorithms and Data Structures (WADS)

- Workshop on Graph-Theoretic Concepts in Computer Science (WG)
- EuroComb and related discrete-mathematics meetings

14 Annotated Source List

Foundational Art-Gallery Theory

Chvatal [7] and Fisk [11].

Foundational $\lfloor n/3 \rfloor$ theorem for simple polygons and the short triangulation/coloring proof.

O’Rourke [20].

Standard book reference for art-gallery theorems, algorithms, visibility, decompositions, and early 3D context.

Shermer [24] and Urrutia [25].

Survey references for art-gallery and illumination problems, useful for terminology, variants, and classical open-problem context.

Abrahamsen–Adamaszek–Miltzow [1].

Modern complexity reference proving $\exists\mathbb{R}$ -completeness for the general art-gallery problem.

2D Orthogonal Polygons and Holes

Kahn–Klawe–Kleitman [15].

Primary source for the tight $\lfloor n/4 \rfloor$ theorem for simple orthogonal polygons.

Gyori [13].

Short proof of the rectilinear art-gallery theorem.

Hoffmann–Kriegel [14].

Graph-coloring result with consequences for rectilinear polygons with holes and prison-yard variants.

Zylinski [31].

Coloring proof of Aggarwal’s theorem for orthogonal galleries with holes, including the $\lfloor (n + h)/4 \rfloor$ bound for $h \leq 2$ and cactus-dual cases.

Urrutia open-problem page [26].

Direct source for Shermer’s and Hoffmann’s open conjectures on orthogonal polygons with holes using vertex guards.

Schuchardt–Hecker [23].

NP-hardness source for art-gallery problems in orthogonal polygons.

Worman–Keil [30].

Decomposition and visibility-graph machinery for the orthogonal art-gallery problem.

Bjørling-Sachs [5].

Tight edge-guard theorem for rectilinear polygons; relevant when comparing 2D orthogonal edge guards with 3D edge-guard conjectures.

Katz–Roisman [16].

NP-hardness for guarding vertices of rectilinear domains by vertex, boundary, or point guards.

Polyomino, Perimeter, and Lattice Variants

Maßberg [17].

Primary peer-reviewed source for perimeter-over-six in hole-free polyominoes.

Maßberg habilitation [18].

Source for Conjecture 6.10 and the conditional hole extension. This is the key source for saying the holes case is not yet proven.

Biedl–Irfan–Iwerks–Kim–Mitchell [3].

Tight $\lfloor (m+1)/3 \rfloor$ area theorem for polyominoes, holes allowed, plus complexity context.

Diaz-Banez et al. [8].

Recent tight theorem for ortho-unit polygons and source of the integral orthogonal polygon $\lfloor N/6 \rfloor$ conjecture.

Pinciu [22].

Polyform extension including polyominoes, polycubes, and polyhypercubes. Relevant for discrete 3D analogues.

Filtser–Krohn–Nilsson–Rieck–Schmidt [10].

Current work on k -hop visibility in polyominoes, including open approximation questions for polyominoes with holes.

3D Orthogonal Polyhedra

Paterson–Yao [21].

Binary-space partition result used for the $\Theta(n^{3/2})$ point-guard bound in orthogonal polyhedra.

Viglietta thesis [27].

Central survey and source for guarding/searching polyhedra, the point-guard obstruction, and Urrutia-type edge-guard conjectures.

Benbernou et al. [2].

CCCG source for improved edge-guard bounds in orthogonal polyhedra: asymptotic $(11/72)m$ and $(7/12)r$.

Viglietta 2-reflex paper [29].

Proves optimal-style reflex-edge bounds for 2-reflex orthogonal polyhedra and gives an $O(n \log n)$ guard-placement algorithm.

Viglietta face-guarding paper [28].

Face-guarding results for several classes of polyhedra, including orthogonal subclasses.

Cano–Toth–Urrutia–Viglietta [6].

General edge-guard result in three-space and explicit open gap.

Algorithms and Related Visibility Models

Motwani–Raghunathan–Saran [19].

Perfect-graph approach to covering orthogonal polygons with star polygons.

Ghosh [12].

General approximation-algorithm context for art-gallery problems.

Durocher et al. [9].

Sliding-camera model for orthogonal art galleries, including holes.

Biedl et al. [4].

Sliding k -transmitter hardness and approximation results for orthogonal polygons.

15 Working Conclusions

1. The theorem “ $\ell/6$ guards suffice” should be cited as proven only for *hole-free polyominoes*. The polyomino-with-holes version is conditional on Maßberg’s rectangle-packing conjecture and should be described as open unless a new proof is supplied.
2. The broad integral statement “an integral orthogonal polygon of perimeter N is guardable by $\lfloor N/6 \rfloor$ guards” is a current conjecture, not a theorem.
3. The right 3D terminology is *orthogonal polyhedron* or *orthogonal polytope*. “Orthonormal” should not be used for the gallery object.
4. A 3D point-guard analogue of the planar perimeter theorem is blocked by the $\Theta(n^{3/2})$ point-guard bound. A serious 3D project should choose edge guards, reflex-edge guards, face guards, or a discrete polycube visibility model explicitly.
5. The strongest project targets suggested by this review are: prove or refute Maßberg’s hole-packing conjecture; prove or refute the integral $\lfloor N/6 \rfloor$ conjecture; or make progress on the Urrutia $m/12 + O(1)$ and reflex-edge conjectures for general orthogonal polyhedra.

References

- [1] M. Abrahamsen, A. Adamaszek, and T. Miltzow. The art gallery problem is $\exists\mathbb{R}$ -complete. *Journal of the ACM*, 69(1), Article 4, 2022. <https://doi.org/10.1145/3486220>
- [2] N. M. Benbernou, E. D. Demaine, M. L. Demaine, A. Kurdia, J. O’Rourke, G. T. Toussaint, J. Urrutia, and G. Viglietta. Edge-guarding orthogonal polyhedra. In *Proceedings of the 23rd Canadian Conference on Computational Geometry*, 461–466, 2011. https://erikdemaine.org/papers/EdgeGuarding_CCCG2011/ <https://www.science.smith.edu/~jorourke/Papers/EdgeGuarding.pdf>
- [3] T. C. Biedl, M. T. Irfan, J. Iwerks, J. Kim, and J. S. B. Mitchell. The art gallery theorem for polyominoes. *Discrete & Computational Geometry*, 48, 711–720, 2012. <https://doi.org/10.1007/s00454-012-9429-1> <https://researchconnect.suny.edu/en/publications/the-art-gallery-theorem-for-polyominoes/>

- [4] T. Biedl, T. M. Chan, S. Lee, S. Mehrabi, F. Montecchiani, H. Vosoughpour, and Z. Yu. Guarding orthogonal art galleries with sliding k -transmitters: hardness and approximation. *Algorithmica*, 81(1), 69–97, 2019. <https://doi.org/10.1007/s00453-018-0433-6>
- [5] I. Björling-Sachs. Edge guards in rectilinear polygons. *Computational Geometry: Theory and Applications*, 11(2), 111–123, 1998. [https://doi.org/10.1016/S0925-7721\(98\)00024-8](https://doi.org/10.1016/S0925-7721(98)00024-8)
- [6] J. Cano, C. D. Toth, J. Urrutia, and G. Viglietta. Edge guards for polyhedra in three-space. *Computational Geometry: Theory and Applications*, 104, 101859, 2022. <https://doi.org/10.1016/j.comgeo.2022.101859> <https://giovanniviglietta.com/papers/edgeguards.pdf>
- [7] V. Chvatal. A combinatorial theorem in plane geometry. *Journal of Combinatorial Theory, Series B*, 18(1), 39–41, 1975. [https://doi.org/10.1016/0095-8956\(75\)90061-1](https://doi.org/10.1016/0095-8956(75)90061-1)
- [8] J. M. Diaz-Banez, P. Horn, M. A. Lopez, N. Marin, A. Ramirez-Vigueras, O. Sole-Pi, A. Stevens, and J. Urrutia. Ortho-unit polygons can be guarded with at most $\lfloor (n - 4)/8 \rfloor$ guards. *Graphs and Combinatorics*, 41, 15, 2025. <https://doi.org/10.1007/s00373-024-02880-8> <https://arxiv.org/abs/2208.12864>
- [9] S. Durocher, O. Filtser, R. Fraser, A. D. Mehrabi, and S. Mehrabi. Guarding orthogonal art galleries with sliding cameras. *Computational Geometry: Theory and Applications*, 65, 12–26, 2017. <https://doi.org/10.1016/j.comgeo.2017.04.001>
- [10] O. Filtser, E. Krohn, B. J. Nilsson, C. Rieck, and C. Schmidt. Guarding polyominoes under k -hop visibility. *Algorithmica*, 87, 572–593, 2025. <https://doi.org/10.1007/s00453-024-01292-7>
- [11] S. Fisk. A short proof of Chvatal’s watchman theorem. *Journal of Combinatorial Theory, Series B*, 24(3), 374, 1978. [https://doi.org/10.1016/0095-8956\(78\)90059-X](https://doi.org/10.1016/0095-8956(78)90059-X)
- [12] S. K. Ghosh. Approximation algorithms for art gallery problems in polygons. *Journal of Discrete Algorithms*, 8(1), 2010. <https://doi.org/10.1016/j.jda.2009.12.004>
- [13] E. Gyori. A short proof of the rectilinear art gallery theorem. *SIAM Journal on Algebraic and Discrete Methods*, 7(3), 452–454, 1986. <https://doi.org/10.1137/0607051>
- [14] F. Hoffmann and K. Kriegel. A graph-coloring result and its consequences for polygon-guarding problems. *SIAM Journal on Discrete Mathematics*, 9(2), 210–224, 1996. <https://doi.org/10.1137/S0895480194265611> <https://refubium.fu-berlin.de/handle/fub188/18382>
- [15] J. Kahn, M. Klawe, and D. Kleitman. Traditional galleries require fewer watchmen. *SIAM Journal on Algebraic and Discrete Methods*, 4(2), 194–206, 1983. <https://doi.org/10.1137/0604020>
- [16] M. J. Katz and G. S. Roisman. On guarding the vertices of rectilinear domains. *Computational Geometry: Theory and Applications*, 39(3), 219–228, 2008. <https://doi.org/10.1016/j.comgeo.2007.02.002>
- [17] J. Maßberg. Perfect graphs and guarding rectilinear art galleries. *Discrete & Computational Geometry*, 51, 569–577, 2014. <https://doi.org/10.1007/s00454-014-9587-4>
- [18] J. Maßberg. Perfect graphs and related topics: habilitation thesis, including Conjecture 6.10. Ulm University, 2013. https://www.uni-ulm.de/fileadmin/website_uni_ulm/mawi.inst.080/Massberg/habilitation_massberg.pdf

- [19] R. Motwani, A. Raghunathan, and H. Saran. Covering orthogonal polygons with star polygons: The perfect graph approach. *Journal of Computer and System Sciences*, 40(1), 19–48, 1990. [https://doi.org/10.1016/0022-0000\(90\)90017-F](https://doi.org/10.1016/0022-0000(90)90017-F)
- [20] J. O’Rourke. *Art Gallery Theorems and Algorithms*. Oxford University Press, 1987. https://www.science.smith.edu/~jorourke/books/ArtGalleryTheorems/Art_Gallery_Full_Book.pdf
- [21] M. S. Paterson and F. F. Yao. Optimal binary space partitions for orthogonal objects. *Journal of Algorithms*, 13(1), 99–113, 1992. [https://doi.org/10.1016/0196-6774\(92\)90007-Y](https://doi.org/10.1016/0196-6774(92)90007-Y)
- [22] V. Pinciu. Guarding polyominoes, polycubes and polyhypercubes. *Electronic Notes in Discrete Mathematics*, 49, 159–166, 2015. <https://doi.org/10.1016/j.endm.2015.06.024>
- [23] D. Schuchardt and H.-D. Hecker. Two NP-hard art-gallery problems for orthogonal polygons. *Mathematical Logic Quarterly*, 41(2), 261–267, 1995. <https://doi.org/10.1002/malq.19950410212>
- [24] T. C. Shermer. Recent results in art galleries. *Proceedings of the IEEE*, 80(9), 1384–1399, 1992. <https://doi.org/10.1109/5.163407>
- [25] J. Urrutia. Art gallery and illumination problems. In J.-R. Sack and J. Urrutia, editors, *Handbook of Computational Geometry*, 973–1027. Elsevier, 2000. <https://doi.org/10.1016/B978-044482537-7/50023-1>
- [26] J. Urrutia. Illumination of polygons: open problems. <https://www.matem.unam.mx/~urrutia/openprob/Polygons/>
- [27] G. Viglietta. Guarding and searching polyhedra. Ph.D. thesis; arXiv:1211.2483, 2012. <https://arxiv.org/abs/1211.2483> <https://giovanniviglietta.com/papers/thesis.pdf>
- [28] G. Viglietta. Face-guarding polyhedra. *Computational Geometry: Theory and Applications*, 47(8), 833–846, 2014. <https://doi.org/10.1016/j.comgeo.2014.04.009>
- [29] G. Viglietta. Optimally guarding 2-reflex orthogonal polyhedra by reflex edge guards. *Computational Geometry: Theory and Applications*, 86, 101589, 2020. <https://doi.org/10.1016/j.comgeo.2019.101589> <https://arxiv.org/abs/1708.05469>
- [30] C. Worman and J. M. Keil. Polygon decomposition and the orthogonal art gallery problem. *International Journal of Computational Geometry & Applications*, 17(2), 105–138, 2007. <https://doi.org/10.1142/S0218195907002264>
- [31] P. Zylinski. Orthogonal art galleries with holes: A coloring proof of Aggarwal’s theorem. *The Electronic Journal of Combinatorics*, 13(1), R20, 2006. <https://doi.org/10.37236/1046> <https://www.combinatorics.org/ojs/index.php/eljc/article/view/v13i1r20>